

Before the Machine

From fundamental fields to the physical requirements of fusion

In the deuterium–tritium fusion reaction we will study, matter changes its nuclear arrangement and releases energy. Before asking how to build a machine that sustains it, we will ask what that matter is, why it holds together, and how its interactions make another arrangement possible. The route begins with the electromagnetic and strong interactions and follows them toward nucleons, nuclei, and fusion.

Our aim is an analytic explanation whose steps we can examine. We will take the specified fundamental theory and its parameters as our starting point. Where we can derive a relation, we will show the argument. Where the complete derivation is missing, we will identify the missing question and the evidence that lets physics proceed. Confinement and the mass gap enter here because they concern the very existence and properties of the objects from which the nuclear story is built.

1 What is inside the fuel

Begin with an atom. Its nucleus contains positively charged protons and electrically neutral neutrons, collectively called *nucleons*. Electrons carry negative electric charge and occupy quantum states around the nucleus. Electromagnetism binds the atom; understanding the nucleus requires the strong interaction as well.

In quantum field theory, particles are states of underlying fields. The electron is elementary in the Standard Model. A nucleon is composite: its description involves quark and gluon fields. Quarks come in different species, called *flavors*; the up and down flavors are the ones needed to label the proton and neutron. Their net valence contents are

$$p : uud, \quad n : udd.$$

These labels specify the excess of quarks over antiquarks of each flavor. The full quantum state also contains gluonic dynamics and quark–antiquark fluctuations. They do not describe three stationary objects joined by rigid rods.^[1]

An up quark has electric charge $+2e/3$ and a down quark $-e/3$, where $e > 0$ is the magnitude of the electron charge. Electric charges add, so

$$q_p = \left(\frac{2}{3} + \frac{2}{3} - \frac{1}{3}\right)e = e, \quad q_n = \left(\frac{2}{3} - \frac{1}{3} - \frac{1}{3}\right)e = 0.$$

This addition has a simple symmetry expression. Under an electromagnetic phase rotation, a field of dimensionless charge Q acquires a factor $e^{iQ\theta}$. A product acquires $e^{iQ_1\theta}e^{iQ_2\theta} = e^{i(Q_1+Q_2)\theta}$. The symmetry is called $U(1)$; the charge assignments specify how the fields transform under it.

The proton’s positive charge already identifies a difficulty for fusion: two positively charged nuclei repel as they approach. Before that encounter can be understood, however, we must explain the other interaction that makes each incoming nucleus possible. Quarks also carry *color*, the charge associated with the strong interaction, described by *quantum chromodynamics* (QCD).

2 What color symmetry explains

A quark's color state is a vector in $\mathbb{C}^3$. Choose the orthonormal basis

$$|r\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |g\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |b\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

conventionally named red, green, and blue. A normalized color state has the form

$$|\psi\rangle = z_r|r\rangle + z_g|g\rangle + z_b|b\rangle, \quad |z_r|^2 + |z_g|^2 + |z_b|^2 = 1,$$

where the coefficients are complex numbers. The transformations are matrices

$$SU(3) = \{U \in \mathbb{C}^{3 \times 3} : U^\dagger U = 1, \det U = 1\}.$$

Unitarity preserves inner products. The condition $\det U = 1$ preserves another useful object: the totally antisymmetric tensor ϵ_{ijk} , called the Levi-Civita symbol. Its components are $+1$ when (i, j, k) is an even permutation of $(1, 2, 3)$, -1 when it is an odd permutation, and 0 whenever an index is repeated. Thus, for example, $\epsilon_{123} = \epsilon_{231} = +1$, whereas $\epsilon_{213} = -1$. Each index runs from 1 to 3, and repeated indices below are summed following Einstein's summation convention.

In the tensor product of three quark color spaces, using the same basis for each quark, consider

$$|C\rangle = \frac{1}{\sqrt{6}} \epsilon_{ijk} |i\rangle \otimes |j\rangle \otimes |k\rangle.$$

There are six nonzero terms, with alternating signs. Applying the same U to the three factors gives

$$\epsilon_{ijk} U^a_i U^b_j U^c_k = (\det U) \epsilon_{abc} = \epsilon_{abc}.$$

Thus $U^{\otimes 3}|C\rangle = |C\rangle$ for every $U \in SU(3)$: applying the same color transformation to all three quarks leaves their combined state unchanged. This is what it means for the state to be color neutral. For fields at different positions, a gauge connection must first transport their color indices to a common point. A quark and an antiquark can also form a singlet because the antiquark transforms in the conjugate representation. These constructions supply the color structure used for ordinary baryons and mesons.^[1]

We have established something exact about representations. The calculation has not determined whether a state binds, what it weighs, or why isolated color is absent from the observed particle spectrum. Those questions require dynamics. Nor does this identity select $SU(3)$ on its own; we have used the gauge group of QCD to calculate a consequence.

How does the field acquire dynamics? A local color transformation can vary with spacetime position x . Differentiating a transformed quark field produces both $U\partial_\mu q$ and $(\partial_\mu U)q$. A gauge field $A_\mu(x)$ supplies the connection needed to compare color at neighboring points. In the convention $D_\mu = \partial_\mu - igA_\mu$, its curvature is

$$[D_\mu, D_\nu] = -igF_{\mu\nu}, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu].$$

Here μ, ν label spacetime directions and g is the strong coupling. The A_μ are matrices, so their commutator need not vanish. The usual gauge-field action is quadratic in F ; this extra term therefore generates interactions among the gluon fields themselves. In the abelian electromagnetic theory the corresponding commutator vanishes.^[2]

The field equations specify interactions. Their quantum dynamics must still explain how those interactions organize into the matter we observe.

3 What it would mean to explain confinement

When quarks are extremely close together, the strong interaction between them becomes weak, and they behave almost like free particles. This is called *asymptotic freedom*.^[2] As they are pulled farther apart, this nearly free behavior gives way to a strong interaction. The contrast with electromagnetism is striking: the electric attraction between opposite charges weakens as they separate, whereas a separating quark and antiquark can remain connected by a narrow tube of gluon field. Stretching this *flux tube* stores more and more energy, rather than allowing the particles to drift apart freely.^[4]

To state a confinement test, first simplify the system. Keep the gluon field and introduce an external, infinitely heavy quark in the fundamental color representation and an antiquark in its conjugate, separated by r . The sources remain fixed while the quantum field responds. We ask how the lowest energy in their presence changes with separation. This is a diagnostic of the theory, rather than a model of a physical proton.

One proposed large-distance behavior is

$$V(r) = V_0 + \sigma r + \text{subleading terms}, \quad F_r = -\frac{dV}{dr} \simeq -\sigma.$$

The positive constant σ is the *string tension*, energy per unit length. The flux-tube picture makes this behavior visible: if field energy stays concentrated in a tube with approximately fixed energy per unit length, stretching it adds energy in proportion to length. Deriving that organization from the quantum theory is the substantive task.

There is a mathematical way to express the energy test. A *Wilson loop* transports color around a closed spacetime path and takes a trace, giving a gauge-invariant quantity. For a rectangle of width r and Euclidean time extent T , ground-state projection gives, with a common source self-energy convention,

$$V(r) = -\lim_{T \rightarrow \infty} \frac{1}{T} \log \langle W(r, T) \rangle.$$

Here $\hbar = c = 1$; Euclidean time is the imaginary-time variable in the continued quantum calculation. If the large rectangle obeys an *area law*,

$$\log \langle W(r, T) \rangle \sim -\sigma r T, \quad \text{then} \quad V(r) \sim \sigma r.$$

This implication identifies what the area law would explain. It does not prove that the continuum theory has that law.^[3]

Now restore dynamical light quarks. At sufficient separation, a state containing two color-singlet heavy-light mesons can be energetically favored over the stretched string. The configurations mix and the string breaks: its energy need not grow forever. Lattice calculations resolve this process. The outgoing objects remain color singlets.^[4]

Confinement in physical QCD must therefore account for the allowed separated states as well as the force between idealized sources. A complete analytic explanation would establish this behavior from a well-defined quantum theory and explain the scales it produces. The observed absence of isolated color and the numerical evidence constrain that explanation; they do not complete it.

4 Why the mass gap is a foundational question

The source experiment asks how energy changes when color sources are separated. A second question concerns the field with no external sources at all: what excitations can exist above its vacuum? The *vacuum* is the lowest-energy quantum state, not a classical field configuration with every fluctuation switched off.

For a free electromagnetic field, a photon of wavelength λ has $E = hc/\lambda$. Arbitrarily long wavelengths permit arbitrarily small positive energies. Classical pure Yang–Mills theory likewise has no explicit gauge-boson mass term. Can its interacting quantum theory instead require a positive minimum excitation energy?

Suppose the theory has been constructed on its physical Hilbert space, with a self-adjoint Hamiltonian H and a unique vacuum $|\Omega\rangle$ in the sector considered. Choose the vacuum energy as zero, so $H|\Omega\rangle = 0$. The *mass-gap condition* is

$$\text{spec}(H) \cap (0, \Delta) = \emptyset \quad \text{for some } \Delta > 0.$$

The spectrum includes discrete eigenvalues and continuous spectral values. The condition excludes every physical excitation with energy between zero and Δ . It asserts a lower bound, not a ladder of equally spaced energies. The positive gap must persist in infinite space and as any lattice spacing goes to zero; level spacing in a finite box does not establish it. A complete formulation must also construct the quantum theory itself. This is the core of the Yang–Mills existence and mass-gap problem in four spacetime dimensions.^[5]

What would such a gap explain? We can derive one consequence conditionally. Let $|\chi\rangle$ be a finite-norm physical state orthogonal to the vacuum, prepared by a suitably smeared gauge-invariant observable. The spectral theorem writes its Euclidean correlation as

$$C(t) = \langle \chi | e^{-tH/\hbar} | \chi \rangle = \int_{\Delta}^{\infty} e^{-Et/\hbar} d\mu_{\chi}(E).$$

The measure is positive and has total weight $\|\chi\|^2$. Since $E \geq \Delta$,

$$0 \leq C(t) \leq e^{-\Delta t/\hbar} \|\chi\|^2 \quad (t \geq 0).$$

A spectral gap thus implies exponential suppression in imaginary time. The missing step is to establish the hypothesized gap for the interacting continuum theory, rather than insert it as the lower integration limit.

In pure Yang–Mills theory, the expected massive color-singlet excitations are called *glueballs*. The gap concerns physical field states; it is not obtained by assigning each gluon an ordinary rest mass. Nor does the spectral inequality alone establish the Wilson-loop area law of Section 3. These are related questions about quantum gauge dynamics with different mathematical targets.^[5]

They belong together in our inquiry: why do the fields restrict which objects can separate, and why do their physical excitations have the energy scales they do?

5 Returning to nucleons and the light pion

Pure Yang–Mills theory isolates the gauge-field question. To reach nucleons we restore the quark fields and their interactions. For nuclear matter we must also include electromagnetism; weak interactions matter for processes that change quark flavor. The next task is to determine the physical states of this richer theory.

A nucleon’s mass measures the rest energy of its interacting quantum state. It is much larger than the sum of the light-quark mass parameters associated with its valence labels. Lattice QCD has reproduced the light-hadron mass spectrum using specified quark masses and a scale-setting input, with controlled numerical extrapolations. This provides substantial evidence connecting QCD to nucleon masses. Our stronger explanatory goal is to understand analytically how that spectrum follows from the dynamics.^[6]

The same distinction applies to size. A nucleon has no rigid material surface; its spatial structure is investigated through responses to probes, described by matrix elements called *form factors*. Masses and form factors ask different questions of the same quantum state. Showing that a singlet is permitted, or that a theory has some positive gap, does not yet calculate either its detailed structure or the full spectrum.^[1]

Why do much lighter particles also appear? Quarks and antiquarks can organize into mesons. The lightest are the pions, with rest energies around 135–140 MeV, compared with about 939 MeV for a nucleon. A megaelectronvolt (MeV) is a million electronvolts; one electronvolt is the energy gained by a charge of magnitude e through a potential difference of one volt. The pion’s small mass brings another property of the QCD vacuum into the explanation: *chiral symmetry breaking*.^{[1][7]}

Set the up- and down-quark masses to zero for this argument and omit electromagnetism. The quark fields have left- and right-chiral components; QCD permits independent flavor rotations in these two sectors. In the accepted symmetry-breaking description, the vacuum retains their common rotations while breaking three relative directions. Goldstone’s theorem then supplies three massless collective excitations, identified with the pions in this limit.

The physical light-quark masses explicitly break that symmetry. At leading order the chiral expansion gives, in natural units,

$$m_\pi^2 = B_\chi(m_u + m_d) + \text{higher-order terms.}$$

Here B_χ encodes a property of the strongly interacting vacuum. The relation explains how small quark masses can give small pion masses once the symmetry-breaking pattern is supplied. It leaves the analytic derivation of that vacuum structure and its coefficients to the underlying theory.^[7]

This is why returning from pure gauge fields to physical QCD is a substantial bridge. Quarks change the spectrum and introduce chiral dynamics. The pion will now carry part of the explanation of how two color-neutral nucleons can interact over a nuclear distance.

6 From the pion to nuclear binding

Color neutrality does not require two nucleons to have zero interaction. Their composite quantum states can exchange energy and momentum. At low energies, one long-range contribution is represented by virtual pion exchange: the pion is internal to the scattering process, while the incoming and outgoing particles can be nucleons. Its coupling and spin structure are inherited from the strong dynamics.^[8]

We can derive the spatial range in a simplified massive-field model. Away from a localized source, a static scalar field satisfies

$$(\nabla^2 - \ell^{-2})\phi = 0, \quad \ell = \frac{\hbar}{mc}.$$

For a spherical solution, let $u(r) = r\phi(r)$. Because $\nabla^2\phi = u''/r$ for $r > 0$, the equation becomes $u'' - u/\ell^2 = 0$. Its solutions are $u = Ce^{-r/\ell} + De^{r/\ell}$. Requiring decay at infinity sets $D = 0$, giving

$$\phi(r) = C \frac{e^{-r/\ell}}{r}.$$

This is the Yukawa range. Using $\hbar c \simeq 197.3 \text{ MeV fm}$ and $m_\pi c^2 \simeq 140 \text{ MeV}$ gives $\ell_\pi \simeq 1.4 \text{ fm}$, where $1 \text{ fm} = 10^{-15} \text{ m}$. The range follows from the stated massive-field equation. The sign and strength of a nucleon interaction do not: actual pion exchange has spin and isospin operators and derivatives, with additional contributions at shorter distances.^[9]

Chiral effective field theory organizes pion exchange, short-range interactions, and forces involving several nucleons according to QCD's symmetries and a low-energy expansion. Many coefficients are determined from data. It supplies a systematic nuclear description while keeping open the stronger task of deriving its quantitative content analytically from QCD.^[8]

Consider the deuteron, the bound proton–neutron pair. With the separated particles at rest assigned zero relative energy, an effective nuclear Hamiltonian has the form

$$H_{\text{rel}} = \frac{\mathbf{p}^2}{2\mu_{pn}} + V_{pn}, \quad \mu_{pn} = \frac{m_p m_n}{m_p + m_n}.$$

A normalizable state satisfying $H_{\text{rel}}\psi = -B_D\psi$, with $B_D > 0$, is bound: energy B_D must be supplied to separate it. This states the binding criterion precisely within the effective description. Finding such a state and its binding energy requires the interaction V_{pn} ; attraction alone does not determine the answer. Quantum localization costs kinetic energy, and the permitted spin and spatial states also matter.

We can now distinguish the bridges we must understand: fields organize into nucleons and pions; their interactions permit nuclei; a reaction connects nuclear arrangements with different energies. Before following the reaction into D–T fusion, we will examine what sets a nucleus's spatial extent and how its binding energy is measured. The analytic gaps in the earlier steps remain part of the inquiry, including how confinement, the vacuum spectrum, and chiral dynamics produce the nuclear interaction and its scales.

7 Why does a nucleus have a finite size?

The binding criterion tells us what to look for: a normalizable state below the separated-particle energy. We now ask why such a state occupies a finite region and what sets its energy. The answer brings quantum motion into the story.

A quantum particle occupies a state spread over space. We describe that state by a wavefunction, a function whose squared magnitude gives the probability density for position. Concentrating that probability into a smaller region requires a wider spread of momenta. Momentum carries kinetic energy, so localizing the particle has an energy cost.

A small calculation makes the competition visible. Look along one coordinate x . Write Δx for the standard deviation of the position distribution and Δp_x for that of the corresponding momentum. These measure the spreads of possible measurement outcomes in the same quantum state. The uncertainty relation is^[10]

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}.$$

Let $L = \Delta x$ be our measure of spatial width. Dividing by L gives $\Delta p_x \geq \hbar/(2L)$.

For a particle of mass m , the nonrelativistic kinetic energy associated with this direction is $p_x^2/(2m)$. Angle brackets denote its quantum average. The definition of variance gives

$$\langle p_x^2 \rangle = (\Delta p_x)^2 + \langle p_x \rangle^2.$$

The smallest cost occurs when the mean momentum is zero. Combining these relations, step by step, gives

$$\langle K_x \rangle = \frac{\langle p_x^2 \rangle}{2m} \geq \frac{(\Delta p_x)^2}{2m} \geq \frac{1}{2m} \left(\frac{\hbar}{2L} \right)^2 = \frac{\hbar^2}{8mL^2}.$$

The important dependence is $1/L^2$: halving the width quadruples this minimum kinetic-energy cost. This one-coordinate calculation isolates the cost of localization; a nuclear calculation also includes the other coordinates, interactions, and the motion of all its constituents.

For a single-nucleon illustration, use a mass $m_N c^2 \simeq 939 \text{ MeV}$ and choose $L = 1 \text{ fm}$. Multiplying numerator and denominator by c^2 lets us use the units already introduced:

$$\frac{\hbar^2}{8m_N L^2} = \frac{(\hbar c)^2}{8(m_N c^2)L^2} \simeq \frac{(197.3 \text{ MeV fm})^2}{8(939 \text{ MeV})(1 \text{ fm})^2} \simeq 5.2 \text{ MeV}.$$

Even this lower bound is on a nuclear energy scale. At half that width it becomes about 21 MeV. For the proton–neutron relative coordinate in Section 6, the mass is instead $\mu_{pn} \simeq m_N/2$; the same one-coordinate width then gives a bound of about 10.4 MeV. Neither illustration predicts the deuteron binding energy without its interaction.

Binding is the outcome of competing contributions. Measure the energy of a proton–neutron pair relative to the same particles separated and at rest. Its internal energy contains a positive kinetic contribution and an interaction contribution that can be negative:

$$E_{\text{relative}} = \langle K_{\text{internal}} \rangle + \langle V_{\text{interaction}} \rangle.$$

Here $V_{\text{interaction}}$ describes their nuclear interaction, and the averages are taken in the same two-particle state.

A very extended state samples little of the attractive region. A very concentrated state pays a high kinetic cost. Some conventional nuclear potentials represent short-distance dynamics by a strongly repulsive core; the detailed form depends on the effective description. Between those extremes, the pair

may find a lower-energy state with a finite spatial spread. A calculation with the actual interaction determines whether that state is bound and by how much.^{[8][9]} This argument identifies contributions to the balance; calculating a nuclear radius and the equilibrium density of nuclear matter still requires the many-nucleon dynamics.

For the deuteron, measurements give $E_{\text{relative}} \simeq -2.225 \text{ MeV}$. It takes 2.225 MeV to separate it into a proton and a neutron with no remaining relative motion. Its *binding energy* is that positive separation cost. Conversely, forming the deuteron from free particles releases energy into other products, such as an emitted photon.^[11]

What changes when we add more nucleons? There are more attractive interactions, but also more states to accommodate. The Pauli exclusion principle says that two identical fermions cannot occupy the same complete one-particle state. Fermions are particles with half-integer spin; protons and neutrons both belong to this class. Protons obey the exclusion rule among themselves, as do neutrons. A state includes both its spatial wavefunction and its *spin*, the particle’s intrinsic angular momentum.^[12]

For a nucleon, spin has two possible projections along any chosen axis. In an independent-particle description, a spatial orbital—one available spatial wavefunction—can therefore hold two protons with opposite spin projections, and likewise two neutrons. As more identical nucleons are added, further spatial orbitals must be used. Their energies and the interactions among all the particles determine the binding of the new nucleus.

We can now see what nuclear binding asks of the physics: find an allowed shared quantum state whose total energy lies below the separated-particle energy. The next step is to compare two different nuclear arrangements. That comparison will give us the energy available from fusion.

8 Reading the reaction before using it

Let Z count protons and N count neutrons; $A = Z + N$ counts nucleons. The symbol A_ZX specifies a nucleus. Changing Z changes the element X ; changing N at fixed Z gives a different isotope. For our fuel and products,

Nucleus	Symbol	Constituents	Electric charge
Deuterium (deuteron)	$D = {}^2_1\text{H}$	$1p + 1n$	$+e$
Tritium (triton)	$T = {}^3_1\text{H}$	$1p + 2n$	$+e$
Helium-4 (alpha)	$\alpha = {}^4_2\text{He}$	$2p + 2n$	$+2e$
Free neutron	n	unbound neutron	0

The principal D–T fusion channel is



Its bookkeeping is $2 + 3 = 4 + 1$ nucleons and $1 + 1 = 2 + 0$ units of charge. The neutron on the right is not inside the helium nucleus. Energy, momentum, and angular momentum must also balance, but these conditions do not force an encounter to fuse: the nuclei can scatter without changing species.^[13]

The Sun’s proton–proton route includes $p + p \rightarrow D + e^+ + \nu_e$, where the weak interaction converts a proton into a neutron during formation of deuterium; e^+ is a positron and ν_e an electron neutrino. Our D–T reaction already has neutron-containing fuel and needs no such weak conversion.^[14]

9 The energy is a difference between arrangements

The deuteron gave us the basic idea: binding lowers the energy of a shared arrangement. We can measure that lowering by weighing the arrangement. Einstein's relation $E_{\text{rest}} = mc^2$ connects the mass of an object to its energy when the object as a whole is at rest. Its internal motion and interactions are included in this rest energy.

Imagine separating a nucleus into its Z protons and N neutrons. Supply just enough energy to leave those particles far apart with no kinetic energy. The energy supplied is the binding energy B . Energy conservation then reads

$$\underbrace{m(Z, N)c^2}_{\text{nucleus at rest}} + B = \underbrace{Zm_p c^2 + Nm_n c^2}_{\text{separated nucleons at rest}}.$$

Here m_p and m_n are the proton and neutron masses, and $m(Z, N)$ is the mass of the nucleus alone; electrons are left out of both sides. Solving for the separation cost gives

$$B(Z, N) = [Zm_p + Nm_n - m(Z, N)]c^2. \quad (2)$$

Thus a bound nucleus weighs less than its separated constituents by B/c^2 . This difference is the *mass defect*. Greater binding means a larger separation cost, and therefore a lower rest energy for that inventory of nucleons.^[13]

Now compare the two arrangements in our reaction. The energy made available by one D–T reaction, called its Q value, is the initial rest energy minus the final rest energy. For each bound nucleus we substitute “separated-nucleon energy minus binding energy.” The free neutron on the right contributes its own rest energy:

$$\begin{aligned} Q &= (m_D + m_T - m_\alpha - m_n)c^2 \\ &= [(m_p + m_n)c^2 - B_D] + [(m_p + 2m_n)c^2 - B_T] \\ &\quad - [(2m_p + 2m_n)c^2 - B_\alpha] - m_n c^2 \\ &= B_\alpha - B_D - B_T. \end{aligned} \quad (3)$$

Both sides contain two protons and three neutrons, so their separate rest energies cancel exactly. What survives is how much more tightly the final arrangement is bound. Evaluated mass data give, rounded to 0.001 MeV, $B_D \simeq 2.225$, $B_T \simeq 8.482$, and $B_\alpha \simeq 28.296$ MeV. Thus

$$Q \simeq 28.296 - 2.225 - 8.482 = 17.589 \text{ MeV} \simeq 17.6 \text{ MeV}.^{[11]}$$

These three binding energies are measured inputs. The calculation shows how they determine the available fusion energy. A microscopic nuclear calculation has a further job: finding the interacting quantum states that reproduce these measured values.

The alpha particle makes the gain possible. In a simple orbital picture, its two protons can occupy one spatial orbital with opposite spin projections, and its two neutrons can do the same. The exclusion rule therefore permits a compact four-nucleon arrangement. Its actual binding follows from the balance of nuclear attraction, quantum motion, and electrical repulsion; the measured 28.296 MeV tells us how favorable that balance is.^{[13][12]}

Where does the released energy go? Into the motion of the alpha particle and neutron. To calculate the sharing, choose a frame in which the total momentum of the incoming pair is zero. This is the *center-of-momentum frame*. Conservation gives

$$\mathbf{p}_\alpha + \mathbf{p}_n = 0, \quad |\mathbf{p}_\alpha| = |\mathbf{p}_n| = p.$$

The products recoil in opposite directions. At the same momentum, the lighter particle moves faster and carries more kinetic energy. Using the nonrelativistic expression $K = p^2/(2m)$ makes the ratio explicit:

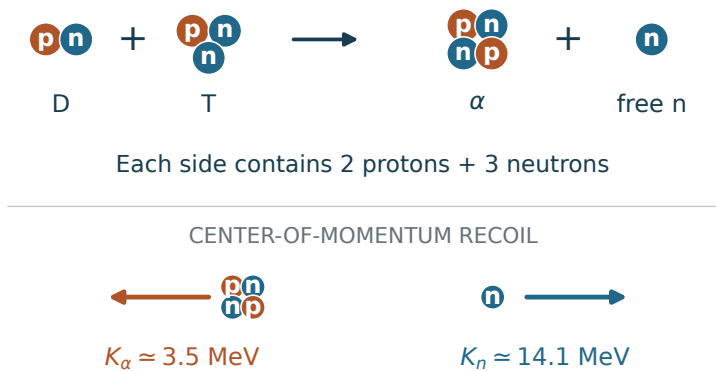
$$\frac{K_n}{K_\alpha} = \frac{p^2/(2m_n)}{p^2/(2m_\alpha)} = \frac{m_\alpha}{m_n} \simeq 4.$$

For incoming kinetic energy small compared with Q , the final kinetic energies add to approximately Q . Substituting $K_n \simeq 4K_\alpha$ gives

$$K_\alpha + K_n \simeq Q \implies 5K_\alpha \simeq Q.$$

Thus $K_\alpha \simeq Q/5 \simeq 3.5 \text{ MeV}$ and $K_n \simeq 4Q/5 \simeq 14.1 \text{ MeV}$. This leading estimate will matter when we build the reactor: most of the released energy travels with a neutral particle. Relativistic two-body kinematics refines these values and includes the incoming kinetic energy.^[16]

A Same nucleons, different arrangement



B Rest-energy difference

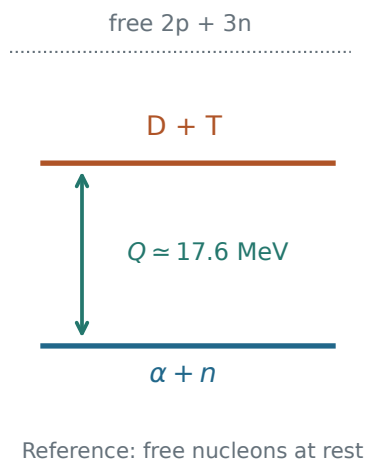


Figure 1: The same $2p+3n$ inventory is rearranged. The two final recoil arrows have equal momentum magnitude, not equal kinetic energy; energies use the leading nonrelativistic estimate in the text. Rest energies are measured relative to separated nucleons. Nucleon circles encode counts, not exact positions or nuclear shapes. Original diagram using sources 11 and 16 and Eq. (3).

10 An energetic downhill route can still have a barrier

The wider pattern is visible by plotting binding energy *per nucleon*, B/A , against nucleon count A . It rises broadly from light nuclei to the iron–nickel region and decreases toward very heavy nuclei. Many light-nucleus fusion reactions and heavy-nucleus fission reactions therefore release energy: their products have greater *total* binding energy than their reactants. For any proposed reaction, Eq. (3)’s method gives the decisive test: compare the total initial and final rest energies.^[13]

Why a peak? The effective nuclear attraction is short ranged: a nucleon mostly benefits from nearby neighbours. Small nuclei have a large surface fraction, so many nucleons have fewer neighbours; larger nuclei initially gain more binding per particle. But electric repulsion acts between every pair of protons, making large charged nuclei increasingly costly. This competition gives the broad trend. Quantum shell structure, proton–neutron imbalance, and pairing determine the detailed curve; this argument alone does not calculate the peak.^[13]

The top of the curve is a broad iron–nickel neighborhood. ^{62}Ni has the largest measured B/A , about 8.795 MeV; ^{56}Fe is nearby at about 8.790 MeV. Stability against a particular decay still depends on its possible products and conserved quantities.^[11]

The energy is downhill; the entrance is uphill. The final arrangement is favorable, but the nucleons must first get close enough to rearrange. A deuteron and triton both carry $+e$. Their electrical repulsion begins far outside the region where nuclear attraction is strong. As they approach, some of their kinetic energy becomes electrical potential energy, slowing the approach.

Let r be the distance between the two nuclei. Treating them as point charges outside their overlap region, Coulomb's law gives the outward force $F_r = e^2/(4\pi\epsilon_0 r^2)$. The constant ϵ_0 is the vacuum permittivity, which sets the electrical-force scale in SI units. The potential energy U satisfies $F_r = -dU/dr$. Choose zero energy at infinite separation and integrate the work needed to push inward:

$$U(r) = \int_r^\infty \frac{e^2}{4\pi\epsilon_0 s^2} ds = \frac{e^2}{4\pi\epsilon_0} \left[-\frac{1}{s} \right]_r^\infty = \frac{e^2}{4\pi\epsilon_0 r}. \quad (4)$$

Using $e^2/(4\pi\epsilon_0) \simeq 1.440 \text{ MeV fm}$, at $r = 5 \text{ fm}$ the cost is

$$U(5 \text{ fm}) \simeq \frac{1.440 \text{ MeV fm}}{5 \text{ fm}} = 0.288 \text{ MeV} = 288 \text{ keV}.^{[15]}$$

Here $1 \text{ keV} = 10^3 \text{ eV}$. For comparison, a head-on encounter with only 10 keV of relative kinetic energy reaches its classical turning point when $U(r) = 10 \text{ keV}$. Solving gives $r \simeq 1.440/0.010 = 144 \text{ fm}$: much farther out than the few-femtometre scale of nuclear attraction. The incoming pair would turn around before reaching the region where fusion can occur.

A quantum wave can reach farther. The wavefunction describes a spread of possible separations. Its behavior inside a barrier follows from the Schrödinger equation. We can see the mechanism with an ordinary differential equation by replacing the curved barrier with a constant height U_0 over a finite interval of width d . Write x for position in this toy model, $E < U_0$ for the conserved energy, and μ for the particle's mass. The stationary equation is^[17]

$$-\frac{\hbar^2}{2\mu} \frac{d^2\psi}{dx^2} + U_0\psi = E\psi.$$

Subtract $U_0\psi$ and multiply by $-2\mu/\hbar^2$:

$$\frac{d^2\psi}{dx^2} = \frac{2\mu(U_0 - E)}{\hbar^2} \psi = \chi^2 \psi, \quad \chi = \frac{\sqrt{2\mu(U_0 - E)}}{\hbar} > 0.$$

The solutions are exponentials, $\psi(x) = Ce^{\chi x} + De^{-\chi x}$. At each edge of the barrier, matching the wavefunction and its derivative joins this solution to the waves outside. A finite transmitted wave remains on the far side. This is *tunnelling*. For a thick barrier the transmitted probability is suppressed approximately by the factor $e^{-2\chi d}$: penetration becomes easier when the barrier is thinner or when the incoming energy is closer to its height.

For the approaching nuclei, the corresponding coordinate is their separation, and its kinetic term uses the reduced mass $\mu = m_D m_T / (m_D + m_T)$. The actual calculation includes the varying Coulomb potential and the possible nuclear outcomes. The toy model isolates the physical opening: a wave can penetrate to small separations even when a classical trajectory would turn back. We still need the probability of fusion once the nuclei interact, which is our next question.

The Coulomb barrier gives a more revealing exponential. For radial motion with zero orbital angular momentum, write $U(r) = a/r$, where $a = e^2/(4\pi\epsilon_0)$. The outer turning point is $r_t = a/E$. The semiclassical WKB approximation adds the local decay exponents across the barrier. The probability of reaching a nuclear matching radius $R < r_t$ contains

$$P_{\text{pen}}(E) \sim \exp \left[-\frac{2}{\hbar} \int_R^{a/E} \sqrt{2\mu \left(\frac{a}{r} - E \right)} dr \right].$$

This treats the exterior Coulomb field as unscreened. To extract its leading energy dependence, formally send the lower limit to zero. This is an analytic idealization, not a claim that nuclei have zero radius. With $r = (a/E) \sin^2 \theta$, the integral becomes

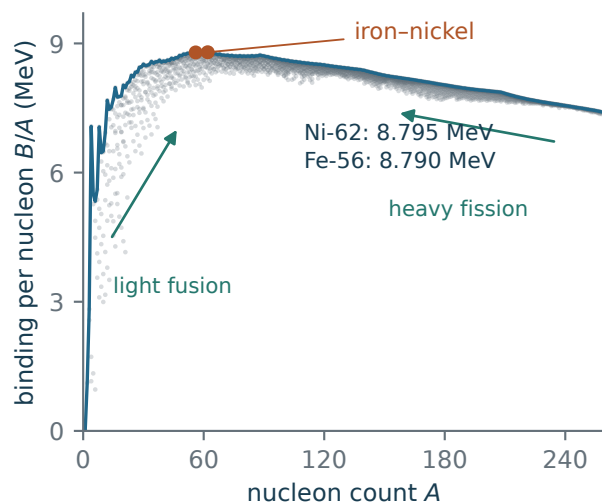
$$a \sqrt{\frac{2\mu}{E}} \int_0^{\pi/2} 2 \cos^2 \theta d\theta = \frac{\pi a}{2} \sqrt{\frac{2\mu}{E}}.$$

Consequently,

$$P_{\text{pen}}(E) \sim e^{-\sqrt{E_G/E}}, \quad E_G = 2\mu \left(\frac{\pi a}{\hbar} \right)^2.$$

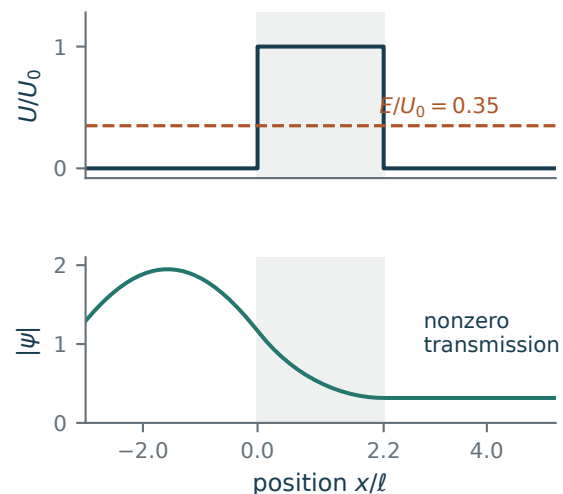
The *Gamow energy* E_G combines electromagnetic strength and reduced mass. This calculation explains the steep suppression at small collision energy. Finite nuclear size, angular momentum, screening where relevant, and the nuclear reaction amplitude still enter a quantitative cross section. Penetration alone does not determine fusion.^[25]

A Binding energy across nuclei



AME2020 measured entries (no # estimates).
Line: largest tabulated B/A at each A .

B An approach barrier



Toy model: $\ell = \hbar/\sqrt{2\mu U_0}$.
Not a D-T cross-section calculation.

Figure 2: Different horizontal axes, different questions. (A) AME2020 entries with measured binding values and $A \leq 260$; estimated values marked # are excluded. The line is an upper envelope of the discrete entries, not a reaction path. Arrows indicate broad possible energy-releasing trends, not that every channel is favorable. (B) An exact stationary solution of the stated constant-barrier toy model: $E/U_0 = 0.35$, width 2.2ℓ , with $\ell = \hbar/\sqrt{2\mu U_0}$. Matching ψ and its derivative gives nonzero transmission; the lower curve is amplitude magnitude, not an energy curve. Original calculation; sources 11 and 17.

11 From a possible reaction to a probability

The symbol $D+T \rightarrow \alpha+n$ identifies an outcome. To predict its frequency, we need a quantum amplitude. A quantum state can be represented by a vector; time evolution changes that vector according to the interactions. For scattering, an operator S connects specified incoming states to outgoing states. In a simplified orthonormal wave-packet basis, write

$$S|i\rangle = \sum_f a_f |f\rangle, \quad a_f = \langle f|S|i\rangle, \quad P_f = |a_f|^2.$$

$|i\rangle$ denotes the prepared incoming state, $|f\rangle$ a resolved outgoing state, and a_f its complex coefficient. Orthogonality gives $\langle f|g\rangle = 0$ for distinct basis states and 1 when $f = g$; probability conservation then reads $\sum_f |a_f|^2 = 1$. Actual continuum momenta require integrals or finite detector bins rather than a probability for one exact momentum.^[18]

An amplitude includes both magnitude and phase. Contributions to the same indistinguishable outcome add *before* squaring; interference can therefore increase or suppress that outcome. We do not select nuclear products by counting nucleons alone.^[18]

How would we measure this? Send a narrow-energy deuteron beam toward a thin target containing N_T tritons. Let Φ_D be the incident number flux, in $\text{m}^{-2}\text{s}^{-1}$, and let $\dot{N}_{\text{fus}}$ count fusion events per second. In the dilute, independent-target limit, the operational definition is

$$\sigma_{DT}(E) = \frac{\dot{N}_{\text{fus}}}{\Phi_D N_T}. \quad (5)$$

Its units are area. It is the reaction's *effective* area at relative collision energy E , not the literal cross-sectional area of a little hard nucleus. The thin-target assumption avoids appreciable beam depletion and energy loss. Connecting an S -matrix element to this area requires incoming flux normalization, outgoing phase-space integration, and appropriate spin sums or averages; one cannot simply set $\sigma = |a|^2$.^[16]

Barrier penetration is only part of the amplitude. D–T fusion is enhanced by a resonance, a temporarily correlated five-nucleon configuration with the quantum numbers of unbound helium-5. A microscopic calculation by Hupin and colleagues treats incoming $D + T$ and outgoing $\alpha + n$ channels with nuclear interactions from effective field theory. Its high-accuracy comparison includes a small phenomenological resonance adjustment. This gives a concrete bridge from a microscopic nuclear model to measured reaction probabilities, with the model's assumptions still visible.^[19]

Where do path integrals enter? An amplitude can also be built by adding contributions from possible histories. Each history carries a phase set by its action divided by $\hbar$. This gives another way to investigate barrier penetration: in a suitable approximation, special paths in imaginary time reveal the tunnelling suppression. Appendix B develops this connection after the main story.^{[20][21]}

12 From one encounter to hot fuel

A reactor needs many reactions, not merely a possible one. Let n_D and n_T denote numbers of deuterons and tritons per unit volume. First imagine all encounters have relative speed v . In a short time dt , one deuteron samples an effective reaction volume $\sigma_{DT}v dt$. That volume contains, on average, $n_T \sigma_{DT}v dt$ tritons. Multiplying by n_D deuterons per unit volume gives the event rate density $\mathcal{R} = n_D n_T \sigma_{DT}v$.

Real particles do not all move at the same speed. Let $g(v) dv$ be the probability that a randomly selected D–T pair has relative speed in $[v, v + dv]$, with $\int_0^\infty g(v) dv = 1$. Averaging the preceding argument gives

$$\langle \sigma v \rangle = \int_0^\infty \sigma_{DT}(v) v g(v) dv, \quad \mathcal{R} = n_D n_T \langle \sigma v \rangle. \quad (6)$$

The brackets mean an average, not multiplication of separately averaged factors. Here $\sigma_{DT}(v)$ is the same cross section, indexed by relative speed rather than collision energy. For unlike species there is no factor 1/2: each D–T pair is counted once. This is a local, dilute, binary-collision model with uncorrelated incoming particles; it is not a formula for an arbitrary strongly correlated state. Dimensional checking gives $[\langle \sigma v \rangle] = \text{m}^3 \text{s}^{-1}$ and $[\mathcal{R}] = \text{m}^{-3} \text{s}^{-1}$. Multiplying by the energy per event gives the local *nuclear* power density, $\mathcal{P}_{\text{fus}} = Q\mathcal{R}$, not net electrical output.

The distribution $g(v)$ is an input to this average. Explaining how electromagnetic forces, collisions, and collective motion establish and evolve it takes us into kinetic and plasma theory. The nuclear cross section supplies one part of that larger dynamical problem.

Deriving the thermal average. Assume independent, isotropic Maxwellian ion distributions at a common temperature T_i , with no relative drift. Write $\Theta_i = k_B T_i$ for the temperature's energy scale. Each Cartesian velocity component is Gaussian with variance Θ_i/m . The difference of the two independent components therefore has variance

$$\text{Var}(v_{D,x} - v_{T,x}) = \frac{\Theta_i}{m_D} + \frac{\Theta_i}{m_T} = \frac{\Theta_i}{\mu}, \quad \mu = \frac{m_D m_T}{m_D + m_T}.$$

The three-dimensional relative Gaussian, integrated over directions, gives

$$g(v) = 4\pi \left(\frac{\mu}{2\pi\Theta_i} \right)^{3/2} v^2 e^{-\mu v^2/(2\Theta_i)}.$$

Substitute this into Eq. (6) and set $E = \mu v^2/2$. Since $v dv = dE/\mu$, we have $v^3 dv = 2E dE/\mu^2$, hence

$$\langle \sigma v \rangle = \sqrt{\frac{8}{\pi\mu}} \Theta_i^{-3/2} \int_0^\infty E \sigma_{DT}(E) e^{-E/\Theta_i} dE.$$

This derives the averaging law from the chosen distributions. It leaves the nuclear cross section, and the conditions maintaining those distributions, to their respective dynamics.^{[19][26]}

Which collision energies contribute? Define $S_{\text{ast}}(E) = E\sigma(E)e^{\sqrt{E_G/E}}$, the *astrophysical factor*, distinct from the scattering operator S . The rate integrand is then proportional to

$$S_{\text{ast}}(E) \exp \left[-\frac{E}{\Theta_i} - \sqrt{\frac{E_G}{E}} \right].$$

High energies are scarce in the thermal distribution; very low energies scarcely penetrate. If S_{ast} varies slowly, minimizing the exponent's positive argument gives

$$\frac{1}{\Theta_i} - \frac{\sqrt{E_G}}{2E^{3/2}} = 0 \quad \implies \quad E_* = \left(\frac{E_G \Theta_i^2}{4} \right)^{1/3}.$$

This estimates the peak within the *Gamow window*, the range of contributing collision energies at a prescribed temperature. For resonant D–T fusion, the full measured or calculated $\sigma_{DT}(E)$ must be used: the slow-variation assumption can fail. The estimate does not select a reactor's optimum temperature.^[26]

To express the thermal scale in familiar units, Boltzmann’s constant gives

$$\Theta_i = 10 \text{ keV} = 10^4 \text{ eV} \quad \Rightarrow \quad T_i = \frac{10^4}{8.617333262 \times 10^{-5}} \text{ K} \simeq 1.16 \times 10^8 \text{ K}.^{[15]}$$

This is a scale illustration, not a temperature optimum derived from the Coulomb barrier. Evaluated cross sections, energy distributions, and losses must be combined to choose an operating range. The familiar multi-keV D–T regime will be justified by those calculations, not assumed to follow from “hotter is better.”^[22]

13 The next bridge: collective plasma dynamics

In the hot-fuel regime used for magnetic fusion, electrons no longer remain bound to individual nuclei: the fuel is an ionized plasma of positive nuclei and negative electrons. It is not a quark–gluon plasma; nucleons remain confined composite particles.^[23] Positive and negative charge can balance overall even though each constituent responds to electromagnetic fields.

Let $f_s(\mathbf{x}, \mathbf{v}, t) d^3x d^3v$ count particles of species s in a phase-space element. In a classical, nonrelativistic mean-field description, the Lorentz acceleration is $\dot{\mathbf{v}} = (q_s/m_s)(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. Its phase-space divergence is zero: $\nabla_{\mathbf{x}} \cdot \mathbf{v} + \nabla_{\mathbf{v}} \cdot \dot{\mathbf{v}} = 0$. Thus the flow preserves phase-space volume. With collisions and sources omitted, particle conservation and the chain rule give

$$\frac{df_s}{dt} = 0 \quad \Rightarrow \quad \frac{\partial f_s}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f_s + \frac{q_s}{m_s} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \nabla_{\mathbf{v}} f_s = 0.$$

This is the Vlasov equation. Maxwell’s equations couple the fields back to the charge and current obtained from the distributions. Collisions and fusion require additional terms.^[27]

The bridge therefore has explicit assumptions: a classical distribution, a collective field description, and a treatment of correlations and collisions. Fluid models take velocity moments and need closures for higher moments. Those steps create further questions about transport and sustained confinement beyond the nuclear reaction rate.^[27]

The first machine question How can we give the fuel enough energetic encounters, and retain that energy long enough, for fusion power to compete with losses? In the magnetic-confinement route pursued here, sustained contact between the hot core and material walls would rapidly remove energy and add impurities. We need separation of core fuel from material surfaces. We have not yet earned a torus or a coil: next we ask whether electromagnetic forces can provide the required control.

The charged alpha particle and neutral neutron will later demand different energy-handling strategies. Microscopic color confinement and macroscopic plasma confinement are different problems; the latter does not follow from the former by analogy.

A Chiral symmetry and the light pion

Section 5 stated the chiral symmetry-breaking pattern and its consequence for the pion mass. The following picture makes the mechanism more intuitive while keeping the dynamical assumption visible.

Picture an energy landscape whose lowest points form a circular trough. Moving radially away from the trough costs energy; moving around its bottom does not. If the system settles at one point on the circle, its chosen state has less symmetry than the landscape: rotating the landscape changes nothing, but rotating the chosen state moves it to a different point. This is the basic idea of *spontaneous symmetry breaking*.

In a field theory, neighboring regions can sit at slightly different points along such a family of equivalent ground states. Slowly varying disturbances then cost little energy. Under the conditions of Goldstone's theorem, an exact continuous symmetry broken by the ground state produces massless collective excitations, called *Goldstone bosons*. The circular trough illustrates one such direction; a theory can have several.^[8]

In QCD with zero up- and down-quark masses, the left- and right-chiral quark components admit independent rotations mixing u and d . The common rotations survive in the accepted vacuum description; three relative directions are broken. The pions are the corresponding collective excitations.^{[8][7]}

The analogy explains the consequence of a symmetry-breaking pattern. Deriving that vacuum structure analytically from QCD remains part of the foundational question.

The physical up and down quarks have small, nonzero masses. These make the chiral symmetry approximate and give the pions a small restoring energy, like a slight tilt making one point of the trough preferable. Pions consequently have nonzero rest energies of about 135–140 MeV. In the massless-quark idealization, with electromagnetism omitted, those pion masses vanish.^{[1][7]}

This connects back to our question about energy gaps. Adding physical light quarks changes the low-energy spectrum; setting their masses exactly to zero is the additional idealization that produces massless pions. For nuclear physics, the measured small-but-positive pion mass gives the finite $\ell_\pi \simeq 1.4$ fm range we used in the main story.

B Tunnelling and instantons

The barrier equation in Section 10 has a complementary path-integral description: which histories contribute to the transition amplitude?

For a particle of mass μ moving in a smooth potential $U(x)$, the *action* of a path is the time integral of kinetic minus potential energy:

$$S[x] = \int \left[\frac{\mu}{2} \left(\frac{dx}{dt} \right)^2 - U(x) \right] dt.$$

The amplitude sums path contributions with phase $\exp(iS[x]/\hbar)$. Action has units of energy times time, so $S/\hbar$ is dimensionless.^[20]

A change of time variable reveals an energy landscape. In problems where the continuation is justified, set $t = -i\tau$, with τ real. Then $dt = -i d\tau$ and $dx/dt = i dx/d\tau$. Substitution gives

$$\begin{aligned} S[x] &= \int \left[-\frac{\mu}{2} \left(\frac{dx}{d\tau} \right)^2 - U(x) \right] (-i d\tau) \\ &= i \int \left[\frac{\mu}{2} \left(\frac{dx}{d\tau} \right)^2 + U(x) \right] d\tau \equiv iS_E[x]. \end{aligned}$$

The quantity S_E is the *Euclidean action*. The phase factor becomes

$$\exp(iS/\hbar) = \exp(-S_E/\hbar).$$

Thus the continued calculation weights paths by an exponential penalty. When the relevant action is large compared with $\hbar$, stationary paths and nearby fluctuations provide a useful *semiclassical approximation*.

For a smooth double-well potential with equal-depth minima, take $U = 0$ at the minima. A finite-action stationary path joining them in imaginary time is an *instanton*. Varying S_E gives

$$\mu \frac{d^2x}{d\tau^2} = \frac{dU}{dx}.$$

The force has the sign of motion in the inverted potential $-U$. An imaginary-time path can therefore connect regions separated by the real-time barrier. Its action sets the leading exponential suppression of the tunnelling contribution.^[21]

For fusion, the paths and boundary conditions must describe incoming nuclei at their collision energy and their nuclear interactions. The double well illustrates the method; Coulomb penetration requires its own semiclassical paths.

An active mathematical direction: Chen’s instanton density. For Yang–Mills fields, instantons can connect vacuum configurations distinguished by a *winding number*, an integer describing topological wrapping. An *instanton density* records local contributions to the corresponding topological charge throughout Euclidean spacetime; integrating it gives the total charge under suitable boundary conditions.^{[21][24]}

Jing-Yuan Chen studies how to preserve this information on a lattice. His construction refines the usual group-valued variables using higher category theory, allowing a local instanton density operator to be defined.^[24] This preserves specified topological information; establishing the continuum dynamics and its spectrum remains a further task. It returns us to the opening’s question of how a representation of quark–gluon physics can retain the mathematical structure of the underlying theory.

Source notes

These references identify the physical inputs and the limits of the arguments. The text develops the color contraction, spectral bound, massive-field range, localization estimate, reaction-energy cancellation, barrier illustration, and rate-counting argument. Each uses stated assumptions; the missing analytic derivation from the underlying quantum fields remains explicit. Section numbers in source descriptions refer to the cited work unless stated otherwise.

[1] Quark content and hadronic states

C. Amsler, V. Crede and T. DeGrand, Particle Data Group, *Quark Model*, 2025 update. See Sections 15.1–15.3, Table 15.1, and Section 15.5 for quark labels, electric charges, mesons, and baryons. The introductory distinction between quark-model classifications and QCD dynamics informs the opening. Charge addition and the three-color antisymmetric contraction are calculated explicitly here; they are not proofs of dynamical confinement.

[2] The gauge field and asymptotic freedom

J. Huston, K. Rabbertz and G. Zanderighi, Particle Data Group, *Quantum Chromodynamics*, 2025 update, Section 9.1, Eqs. (9.1)–(9.3). These specify the QCD Lagrangian, gluon self-interactions, and perturbative running. The opening uses the matrix convention $D_\mu = \partial_\mu - igA_\mu$ and defines F by its commutator; signs in component conventions may consequently differ. The gauge-field action and quantum framework are supplied physical theory, not derived here from group theory alone.

[3] Wilson loops and the confinement diagnostic

K. G. Wilson, *Confinement of Quarks*, Physical Review D 10, 2445–2459 (1974), Sections II and IV. The static-source energy relation assumes a valid spectral projection and consistent treatment of source self-energies. Its elementary large- T consequence is shown in the text. Wilson’s strong-coupling lattice argument identifies an area law in that regime; establishing the required continuum limit remains a separate task.

[4] Dynamical string breaking

G. S. Bali et al., *Observation of String Breaking in QCD*, Physical Review D 71, 114513 (2005). See the abstract and the mixing of the static quark–antiquark and two static–light-meson sectors. This is numerical evidence for the behavior with dynamical light quarks. It motivates treating the unbroken flux tube and the broken-string configuration as distinct, mixing descriptions, rather than extending a linear potential indefinitely.

[5] Existence and the spectral gap

A. Jaffe and E. Witten, *Quantum Yang–Mills Theory*, Sections 4–5, printed pp. 6–7. Section 4 states the existence and mass-gap problem for compact simple gauge groups in four dimensions; Section 5 discusses consequences and further questions, including confinement and additional fields. The [Clay problem page](#) continues to list it as unsolved. The temporal correlation bound in the opening is an application of the spectral theorem assuming a constructed theory and a gap. It is not a proof of either assumption, nor a claim that pure Yang–Mills theory already contains nucleons.

[6] Numerical evidence for hadron masses

S. Dürr et al., *Ab-initio Determination of Light Hadron Masses*, Science 322, 1224–1227 (2008), open manuscript. The calculation tests the light-hadron spectrum with controlled lattice extrapolations and physical input parameters. The paper’s term *ab initio* names its numerical QCD starting point. This opening separately states its aspiration to an analytic explanation; the terminology does not erase the distinction.

[7] Chiral symmetry and the pion

H. Leutwyler, *Light Quark Effective Theory* (1996), especially Sections 1 and 3 and Eq. (3). The leading dependence of the pion mass on the light-quark masses is used with electromagnetism omitted and a common renormalization convention understood. See also Machleidt and Entem, source 8, Section 2.1, for the symmetry-breaking pattern. The opening explains the consequence of that pattern and leaves its full analytic derivation from QCD explicit as part of the deeper task.

[8] The bridge to nuclear interactions

R. Machleidt and D. R. Entem, *Chiral Effective Field Theory and Nuclear Forces*, Physics Reports 503, 1–75 (2011). See the introductory account of QCD and nuclear forces, Section 2 on chiral symmetry, and the treatment of pion exchange, contact terms, and many-nucleon forces. For a detailed account of fitted coefficients and uncertainty, see E. Epelbaum, H. Krebs and P. Reinert, *High-Precision Nuclear Forces From Chiral EFT*, Frontiers in Physics 8, 98 (2020), Sections 2 and 7. The effective Hamiltonian in Section 6 is a declared nuclear description; no particular potential or predicted deuteron binding energy is supplied.

[9] Exponential range and the nuclear potential.

R. Machleidt, “Nuclear Forces,” *Scholarpedia* 9(1), 30710 (2014), subsection “Yukawa’s idea of 1935,” derives the exponential-over-distance form from the massive-field equation. RIKEN’s first-party account of the lattice calculation by N. Ishii, S. Aoki, and T. Hatsuda, “Don’t Get Too Close” (2007), explains the attractive region, repulsive core, and exponentially decreasing tail. The central-potential picture is a useful description of the competition; detailed calculations include spin and other operator structure.

[10] Localization, momentum spread, and kinetic energy.

Feynman, Leighton, and Sands, Vol. III, Chapter 16: Dependence of Amplitudes on Position, especially Eq. (16.35) for $\Delta x \Delta p_x \geq \hbar/2$, and Chapter 20: Operators. Section 7 combines this uncertainty relation with the variance identity and nonrelativistic kinetic energy; the 5.2 MeV is a one-coordinate localization bound for the stated width.

[11] Numerical binding-energy inputs.

Atomic Mass Evaluation 2020, [mass_1.mas20](#) from the IAEA Atomic Mass Data Center. Rows $(Z, A) = (1, 2), (1, 3), (2, 4), (26, 56), (28, 62)$ provide the relevant values; the binding column is *keV per nucleon*, so multiply by A and divide by 1000 for total MeV. Atomic-mass conventions require electronic-binding corrections for high-precision bare-nucleus energies; they are immaterial to the 0.001 MeV inputs used here.

[12] Identical nucleons, spin, and the exclusion principle.

Feynman, Leighton, and Sands, Vol. III, Chapter 4: Identical Particles, particularly §4–2 on the Pauli principle and §4–3 on nuclear scattering and spin. The orbital explanation in the opening is an independent-particle guide to allowed occupations; the many-body interaction sets the binding energy.

[13] Nuclear notation and the binding-energy trend.

P. Cappellaro, MIT 22.02, Chapter 1: Introduction to Nuclear Physics (2012), §1.1.1 and §1.2. The liquid-drop explanation is qualitative here; no fitted mass formula is used to predict light-nucleus binding energies or the exact peak location.

[14] The deliberately short weak-interaction detour.

L. E. Marcucci, R. Schiavilla, and M. Viviani, “The Proton–Proton Weak Capture in Chiral Effective Field Theory” (2013). The solar reaction is not free proton decay, nor the only physical route by which deuterium can form.

[15] Units and temperature conversion.

NIST, SI Brochure, §2: The International System of Units. Exact defining values: $e = 1.602176634 \times 10^{-19}$ C and $k_B = 1.380649 \times 10^{-23}$ J K^{−1}. Their quotient gives $k_B = 8.617333262 \dots \times 10^{-5}$ eV K^{−1}. Together with NIST’s CODATA table entry for vacuum electric permittivity, these give $e^2/(4\pi\epsilon_0) \simeq 1.440$ MeV fm.

[16] Two-body kinematics and the meaning of cross section.

Particle Data Group (2025), “Kinematics,” §49.3 (invariant amplitudes), §49.4.2 (two-body final states), and §49.5 (cross sections). The main-text recoil result is a leading nonrelativistic estimate, not an exact relativistic energy partition.

[17] The quantum law used in the barrier illustration.

R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, Vol. III, Chapter 16, especially

the Schrödinger-equation discussion. Our constant-barrier ODE illustrates penetration only; it is not used as a numerical D–T cross-section model.

[18] Amplitudes and interference.

Feynman, Leighton, and Sands, [Vol. III, Chapter 3: Probability Amplitudes](#), especially the rules for combining amplitudes. The main text uses a normalized wave-packet basis to avoid assigning a finite probability to one exactly specified continuum momentum.

[19] A real microscopic D–T calculation and its limits.

G. Hupin, S. Quaglioni, and P. Navrátil, “[Ab Initio Predictions for Polarized Deuterium–Tritium Thermonuclear Fusion](#),” *Nature Communications* **10**, 351 (2019). See Results, Fig. 1, and Methods: the “NCSMC-pheno” result includes a -5 keV resonance centroid correction. No polarized-fuel enhancement is assumed in our opening.

[20] Action and the sum over histories.

Feynman, Leighton, and Sands, [Vol. II, Chapter 19: The Principle of Least Action](#), concluding quantum discussion. This supplies an accessible conceptual entry, not a construction of the QCD functional measure.

[21] Imaginary-time paths and quantum-mechanical instantons.

D. Tong, [Gauge Theory, Chapter 2](#), printed pp. 56–59 on quantum-mechanical tunnelling in a double well. It develops the Euclidean path integral, the instanton solution, and its semiclassical interpretation. Appendix B shows the time-continuation algebra; prefactors require the fluctuations about the stationary path.

[22] Later engineering connection.

J. P. Freidberg, F. J. Mangiarotti, and J. Minervini, “[Designing a Tokamak Fusion Reactor—How Does Plasma Physics Fit In?](#)” (2015), MIT report PSFC/JA-15-21. This is the subsequent design target, not the source of a confinement proof or a universal temperature threshold.

[23] Constituents and the meaning of quark–gluon plasma.

US Department of Energy, “[DOE Explains: Quarks and Gluons](#).” Introductory orientation for quark content, gluons, and the distinction between intact nucleons and deconfined matter. The electron-cloud description does not assume classical orbital trajectories.

[24] Preserving instanton topology on a lattice.

J.-Y. Chen, “[Instanton Density Operator in Lattice QCD from Higher Category Theory](#),” *SciPost Physics* **19**, 158 (2025). See §1.1 for the topology problem and §4.2 for the refined lattice Yang–Mills construction.

[25] Coulomb penetration and the Gamow exponent

A. I. Chugunov, H. E. DeWitt and D. G. Yakovlev, [Coulomb Tunneling for Fusion Reactions in Dense Matter: Path Integral Monte Carlo versus Mean Field](#), *Physical Review D* **76**, 025028 (2007), Section IV.B, Eqs. (16)–(17). The opening uses the unscreened limit and zero-angular-momentum radial barrier. The lower-limit idealization extracts the leading Coulomb exponent; it does not supply a nuclear transition amplitude.

[26] Thermal reaction rates and the Gamow window

T. Rauscher, [Relevant Energy Ranges for Astrophysical Reaction Rates](#), *Physical Review C* **81**, 045807 (2010), Eqs. (1)–(5), (8), and Section II.B. These distinguish the Maxwellian average, the astrophysical factor, and the limitations of the usual peak estimate. Source 19 also gives the thermal integral specifically for D–T. The Gaussian relative-velocity argument and change of integration variable are shown in Section 12 of this prologue.

[27] From particle trajectories to a kinetic equation

MIT 16.55, [Ionized Gases, Lecture 8: Kinetic Theory](#) (2014), pp. 1–4: distribution functions, moments, phase-space conservation, and Vlasov and collision terms. Section 13 states the kinetic approximation and derives its collisionless transport equation; it does not derive a closure or confinement law.